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Data-Driven and Physics-Guided Design of Viscosity-Modifying Polymers

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Keywords

shear thinning, Newtonian, rheology, architecture, topology, Bayesian optimization

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Abstract

Modifying solution viscosity is a key functional application of polymers, yet the interplay of molecular chemistry, polymer architecture, and intermolecular interactions makes tailoring precise rheological responses challenging. We introduce a computational framework coupling topology-aware generative machine learning, Gaussian process modeling, and multiparticle collision dynamics to design polymers yielding prescribed shear-rate-dependent viscosity profiles. Targeting thirty rheological profiles of varying difficulty, Bayesian optimization identifies polymers that satisfy all low- and most medium-difficulty targets by modifying topology and solvophobicity, with other variables fixed. In these regimes, we find and explain design degeneracy, where distinct polymers produce near-identical rheological profiles. However, satisfying high-difficulty targets requires extrapolation beyond the initial constrained design space; this is rationally guided by physical scaling theories. This integrated framework establishes a data-driven yet mechanistic route to rational polymer design.

1 Introduction

Tuning the rheological properties of solutions by addition of polymeric materials is critical across diverse technological and industrial applications. In enhanced oil recovery, water-soluble polymers mitigate viscous fingering phenomena, improving oil displacement efficiency and recovery yields.¹⁻³ Within food science, hydrophilic polymers modulate texture, stability, and sensory perception through their flow behavior.^{4,5} The essential target across these applications is to control the shear-rate-dependent viscosity, which depends on a complex interplay between polymer architecture, monomer chemistry, and solvent quality.^{6,7} While extensive research has elucidated structure-property relationships for specific polymer families,⁸⁻¹⁵ predicting solution rheology remains challenging when simultaneously considering diverse polymer architectures and polymer-solvent interactions.

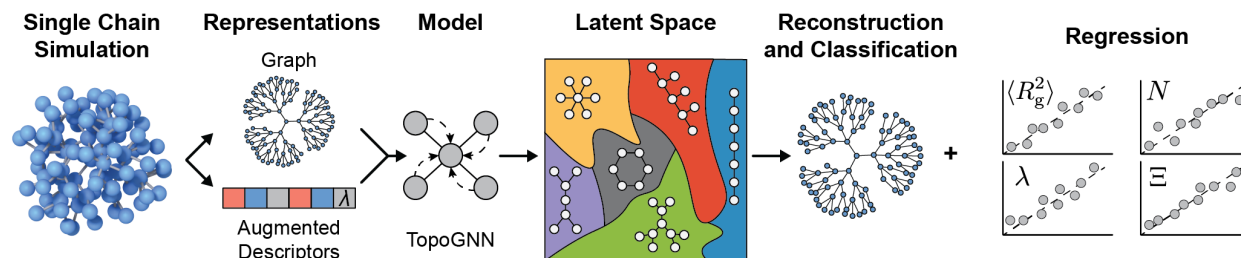
Computational approaches to polymer rheology must balance molecular details with the ability to access relevant timescales and capture key physics. While continuum methods efficiently model macroscopic flow, they cannot resolve molecular mechanisms underlying rheological response.^{16,17} Conversely, atomistic simulations provide molecular detail but are generally limited to spatiotemporal scales that fall short of characterizing the behavior of polymers in flow.¹⁸⁻²² Many coarse-grained particle-based approaches extend accessible timescales, but implicit-solvent treatments typically neglect hydrodynamic interactions, which may yield incorrect or misleading characterizations of polymer rheology, particularly in flow.^{23,24} Multiparticle collision dynamics (MPCD) addresses these limitations by explicitly modeling solvent particles while using simplified collision rules, thereby capturing hydrodynamic correlations essential for rheological prediction at computationally accessible scales.²⁵⁻³⁰ MPCD studies have successfully characterized flow behavior in various polymer architectures,³¹⁻³⁶ establishing it as an effective method to study rheological behavior with a balance of chemical specificity and physical fidelity.

Active learning provides a principled framework to accelerate materials design by iteratively selecting candidates that maximize information gain or performance improvement.

Through Bayesian optimization, active learning has enhanced polymer discovery for sequence optimization, structural design, and property prediction.^{37–47} However, relative to sequence or composition spaces,^{37,43,48,49} applying active learning to topologically complex polymers is challenging because the space of possible polymer topologies cannot be reasonably enumerated. Recently, we mitigated this issue through the development of TopoGNN, a graph-based, physics-informed variational autoencoder that encodes polymers spanning linear, cyclic, branched, star, and dendritic architectures into a continuous latent space.²⁴ This universal representation previously enabled systematic comparison of how different topologies affect concentration-dependent zero-shear viscosity while controlling for dilute-regime behavior.²⁴ More broadly, it allows optimization across multiple architectural classes simultaneously, transcending conventional single-architecture parameter studies.^{50–57}

Here, we present a data-driven framework to design polymer solutions that exhibit targeted shear-rate-dependent viscosity profiles. Our approach integrates three components: graph-based generative modeling via TopoGNN, mesoscale rheological characterization through MPCD simulations, and active learning with Gaussian process surrogates to efficiently explore the topology-solvophobicity design space (Fig. 1). Unlike prior polymer machine learning studies targeting scalar equilibrium properties,^{24,49,58–66} this inverse design of complete rheological curves requires optimization that leverages large-scale non-equilibrium simulations across multiple shear rates per sample point. We demonstrate that our data-driven search successfully identifies diverse polymers to satisfy most target profiles. When active learning reaches its limits for extreme rheological targets, we show how scaling relations enable systematic extrapolation through molecular weight and concentration adjustments. Further analysis elucidates the primary physical features that enable viscosity control. This altogether illustrates how data-driven modeling and physics-based analysis synergistically enable polymer design and understanding of complex polymer properties, with immediate application to solution rheology.

I. Latent Representation Construction



II. Active Learning for Rheology

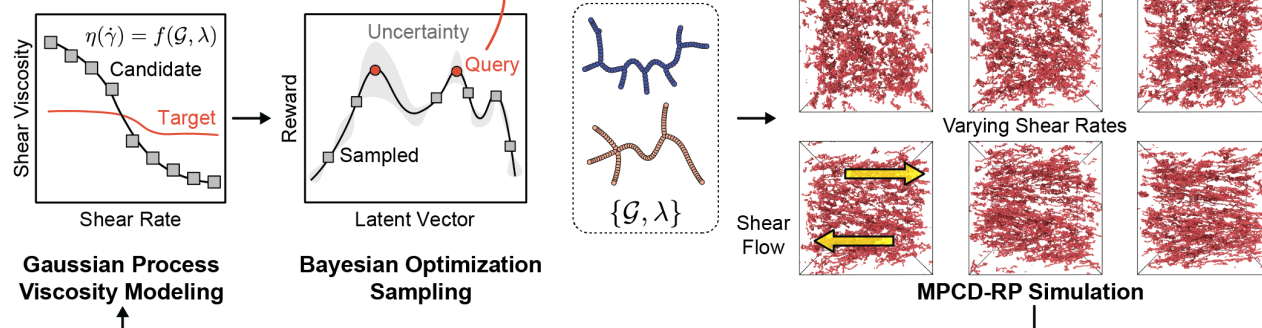


Fig. 1: Data-driven framework for designing viscosity-modifying polymers. The framework operates in two phases. Phase I constructs a continuous latent representation of polymer topology and solvophobicity using a variational autoencoder trained on properties from inexpensive equilibrium simulations: mean squared radius of gyration $\langle R_g^2 \rangle$, monomer count N , solvophobicity λ , chain virial number Ξ , and architectural class. Phase II employs this representation for active learning of rheological targets. Candidate polymers sampled from the latent space undergo non-equilibrium MPCD simulations with reverse perturbation to obtain shear-dependent viscosity profiles $\eta(\dot{\gamma})$. A Gaussian process surrogate maps latent vectors to rheological responses, while Bayesian optimization guides sampling toward underexplored or high-performing regions.

2 Results

Our central objective is to design polymers that exhibit prescribed rheological response profiles. To assess robustness, we define 30 viscosity profiles of varying difficulty and optimize polymer topology and solvophobicity within constraints of fixed concentration, molecular weight, and a bounded solvophobicity range. As outlined in the workflow of Fig. 1, we apply a data-driven design strategy that operates in two phases. Phase I, described in Section 2.1, constructs a universal latent representation from inexpensive equilibrium simulations, encoding both polymer topology and solvophobicity to enable systematic exploration of the design space. Phase II, detailed in Section 2.2, employs active learning to identify can-

didate polymers that aim to satisfy given rheological targets. During each iteration of active learning, the rheological profiles of these candidates are explicitly evaluated using MPCD simulations across a range of shear rates. In Section 2.3, we apply scaling relations from polymer physics to extend the design space and overcome limitations imposed by molecular weight and concentration. Finally, in Section 2.4, we analyze polymer conformations under shear and identify key features that govern viscosity across our design space.

2.1 Class-balanced latent-space sampling captures rich rheological diversity

To explore how polymer topology and solvophobicity jointly influence rheological behavior, we first devise a means to generally represent the polymers by extending our prior generative model TopoGNN. TopoGNN, which is based on a variational autoencoder framework, was initially trained on a dataset, TopoRg-1342, of mean squared radius of gyration $\langle R_g^2 \rangle$ data for 1,342 polymers across six architectural classes: linear, cyclic, branch, comb, star, and dendrimer.²⁴ This yielded a low-dimensional representation of polymers that was sensitive to polymer topology, with some organization by polymer size.

Here, we aim to augment the TopoGNN representation to also be sensitive to monomer solvophobicity λ . To do so, each polymer topology from TopoRg-1342 was modeled with five values of $\lambda \in [0, 0.5]$, with the maximum set to ensure all systems remain in good-solvent conditions (Supplementary Information Section S2). Then, to effectively organize the joint topology-solvophobicity space, the primary reconstruction task of the autoencoder was augmented with four properties: $\langle R_g^2 \rangle$ at $\lambda = 0$, bead count N , solvophobicity λ , and chain virial number Ξ , which together capture structural and solvation effects that influence rheological behavior. Model performance metrics are detailed in Supplementary Information Section S3.2.

The learned latent space provides a computationally efficient foundation for exploring rheological diversity without relying on non-equilibrium flow simulations, which are sub-

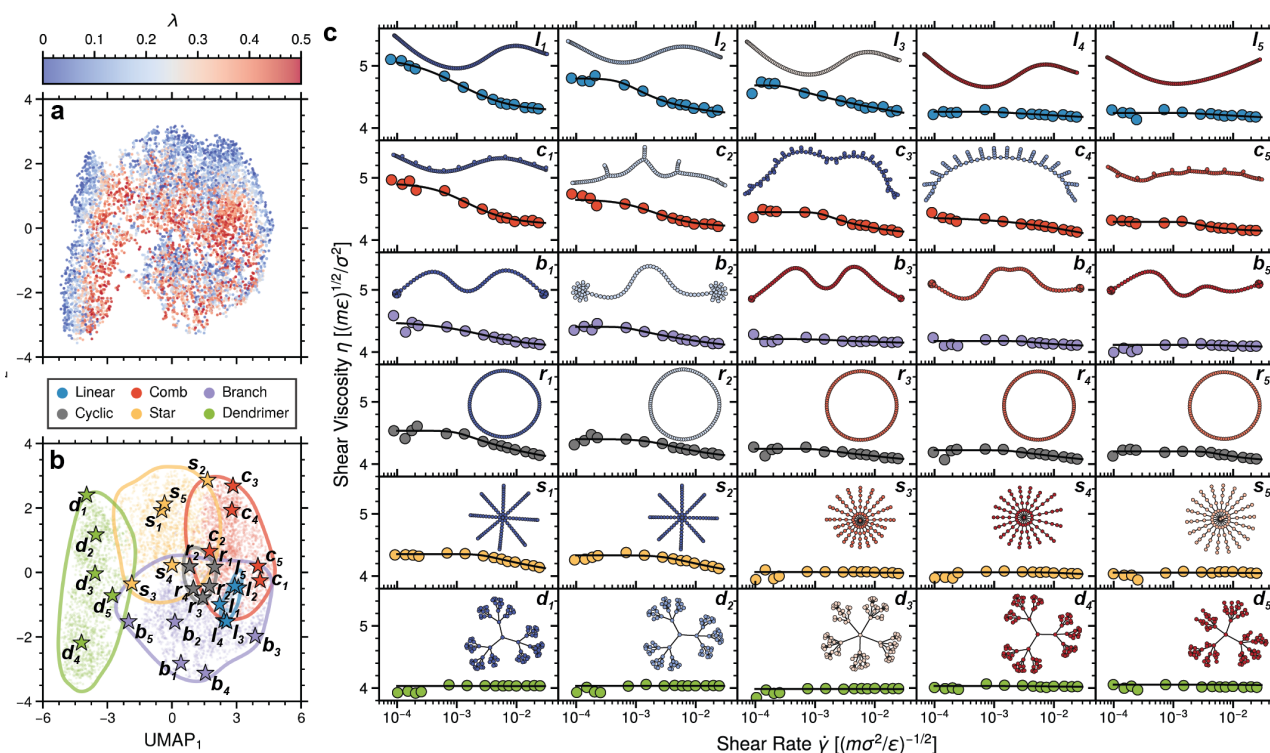


Fig. 2: Latent space structure and rheological properties of seed polymers. **a**, UMAP projection of the latent space learned by `TopoGNN`, with points colored by solvophobicity λ . **b**, UMAP projection annotated with architectural clusters, labeled by polymer architecture: linear (l_i), comb (c_i), branch (b_i), cyclic (r_i), star (s_i), and dendrimer (d_i), where $i = 1-5$ denotes representative instances. Star markers indicate selected seed polymers spanning diverse latent regions. **c**, Shear viscosity profiles of the selected seed polymers. Each row represents a distinct architectural class with its polymer graph; node color encodes λ . Markers denote simulation data; black lines show Carreau–Yasuda fits.

stantially time- and resource-intensive (e.g., 48 GPU hours versus one CPU hour for equilibrium simulations). A two-dimensional projection generated using Uniform Manifold Approximation and Projection (UMAP), which is a nonlinear dimensionality reduction technique, reveals an organized structure with radial gradients in solvophobicity λ and distinct architectural clusters (Fig. 2a,b). Solvophobic polymers concentrate toward the center, while architectural classes separate based on structural features. Dendrimers form tight clusters reflecting their recursive branching, comb and star polymers partially overlap due to shared arm-like motifs, and linear and cyclic chains occupy adjacent regions distinguished by chain connectivity. This structured organization suggests that even simple sampling strategies might capture substantial rheological variation.

To test this hypothesis, we selected 30 seed polymers using class-balanced maximin sampling, ensuring representation across both topology and solvophobicity dimensions without prior knowledge of rheological behavior. Remarkably, this selection, which is computationally inexpensive relative to the full rheological characterization, reveals rich shear-dependent responses spanning nearly two orders of magnitude in viscosity (Fig. 2c). Linear and comb polymers exhibit strong shear-thinning, with viscosity dropping substantially as shear rate increases. Dendrimers maintain Newtonian behavior across all solvophobicity values, while star, cyclic, and branch polymers show intermediate responses.

The diversity arises from an interplay between topological constraints and solvation effects. In flexible topologies, low solvophobicity promotes chain expansion that facilitates flow-induced alignment and viscosity reduction. High solvophobicity drives chain collapse, creating compact structures that resist reorganization under shear. Sterically hindered architectures such as dendrimers remain insensitive to solvophobicity because their structural constraints dominate. This baseline characterization demonstrates that topology sets mechanical limits while solvophobicity modulates the response within those limits. In the following sections, we examine the extent to which active learning facilitates fine-tuning of rheological response through the tandem design of polymer topology and solvophobicity.

2.2 Active learning improves target matching while exposing design space boundaries

To explore the capabilities of active learning-guided design in this space, we first define 30 distinct shear-rate-dependent viscosity profiles that represent targets for optimization. These targets are categorized into three difficulty levels based on their distance from profiles observed in the seed dataset (Fig. 3a). Low-difficulty targets closely resemble seed profiles and likely require only interpolation within the existing design space. Medium-difficulty targets demand moderate extrapolation beyond observed behaviors. High-difficulty targets exhibit viscosity ranges and shear-rate dependencies unlike any polymer in the seed set,

pushing the boundaries of the accessible design space. Each category contains ten targets with systematic variation in zero-shear viscosity, shear-thinning onset, and power-law scaling, as represented in the Carreau-Yasuda model (Supplementary Information, Section S4).

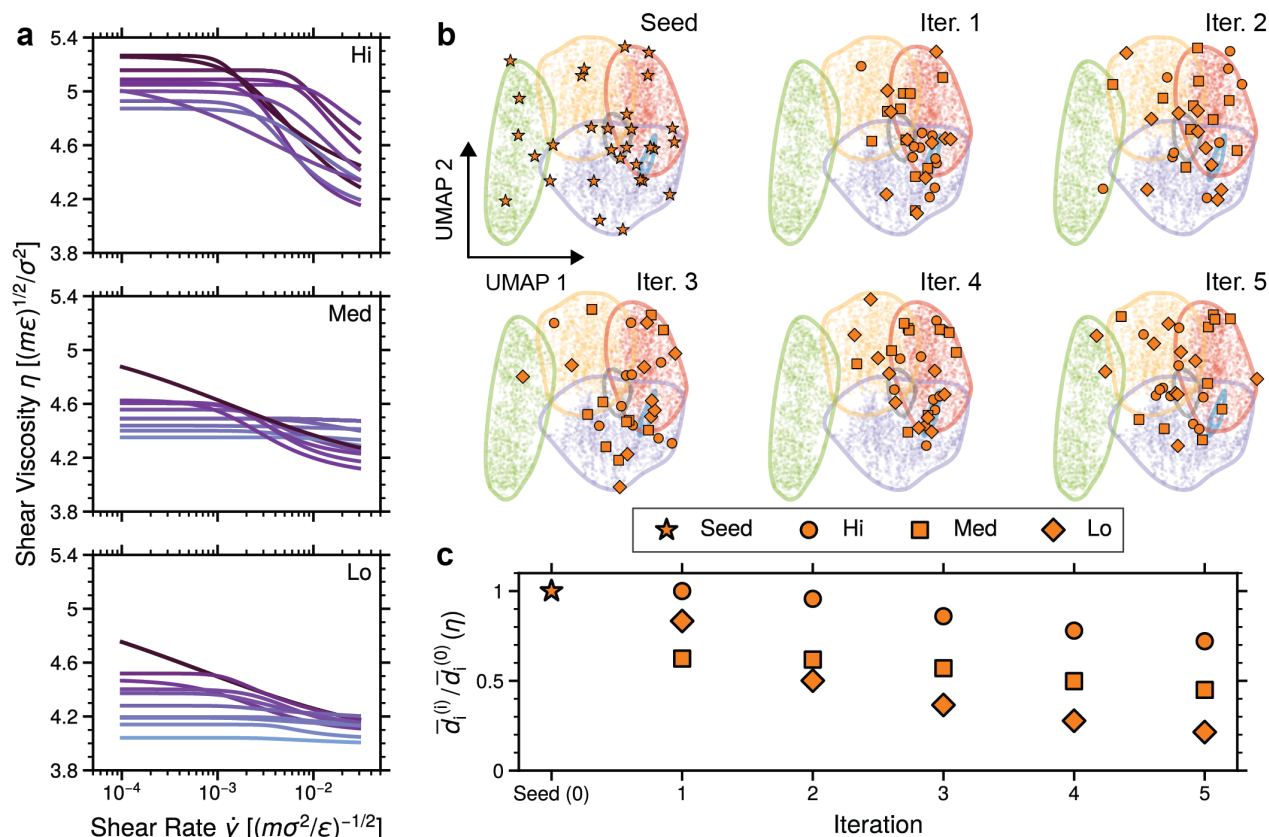


Fig. 3: Active learning performance and procession through latent space. **a**, Target shear viscosity profiles at low-, medium-, and high-difficulty levels; curves are colored by zero-shear viscosity magnitude to visually distinguish curves within a difficulty level. **b**, UMAP projections of latent representations at each active learning iteration. Colored regions indicate regions spanned by architectural classes, as in Fig. 2b. **c**, Normalized average deviation between the top- i discovered and target viscosity profiles at each iteration (i).

Active learning successfully improves match quality across iterations for all target categories. The algorithm explores diverse regions of the latent space, with candidates distributed broadly across architectural clusters (Fig. 3b). Notably, sparse sampling occurs in the dendrimer region despite its substantial representation, likely reflecting the limited rheological variability of polymers with these compact structures. Quantitative analysis shows monotonic improvement in matching performance over five iterations (Fig. 3c). Low-

difficulty targets converge rapidly from an already close starting point ($\bar{d}_0^{(0)} = 3.95 \times 10^{-4}$), while high-difficulty targets begin far from achievable profiles ($\bar{d}_0^{(0)} = 1.04 \times 10^{-1}$) and improve more gradually.

Although active learning is routinely employed for molecular design, its advantage over simpler search strategies often goes unexamined. We therefore compared active learning performance against two space-filling baselines (Supplementary Information, Fig. S5). Active learning outperforms naive space-filling but achieves comparable results to class-balanced sampling that ensures topological diversity. This parity stems from unexpected cross-target generalization revealed by tracking candidate origins (Supplementary Information, Fig. S4). In particular, many high-performing polymers for any given target are found from searches aimed at different profiles. While low- and medium-difficulty targets benefit from candidates discovered while pursuing rheologically similar objectives, high-difficulty targets draw solutions from more distant, exploration-based searches. This may also explain the competitiveness of class-balanced sampling. When architectural diversity naturally yields polymers suitable for multiple targets, the adaptive advantage of active learning diminishes, especially in densely sampled target spaces.

To better appreciate the quality of reproducing target profiles, we compare representative examples of targets and their best solutions; similar results for the remaining targets are provided in the Supplementary Information, Section S10. This illustrates how solution quality and diversity vary with target difficulty. For low-difficulty targets, active learning identifies multiple polymers that closely reproduce the desired viscosity profiles (Fig. 4a). Interestingly, the top five candidates span different topologies and solvophobicity values (Fig. 4b), demonstrating substantial design degeneracy. Medium-difficulty targets are also well reproduced (Fig. 4c), with successful matches requiring more specific combinations of topology and solvophobicity (Fig. 4d). Finally, high-difficulty targets highlight the limits of our constrained design space (Fig. 4e), as targets are generally not satisfactorily captured by any proposed polymers. In particular, active learning converges on linear, solvophilic polymers (Fig. 4f) that partially capture target features but fail to achieve the extreme shear-

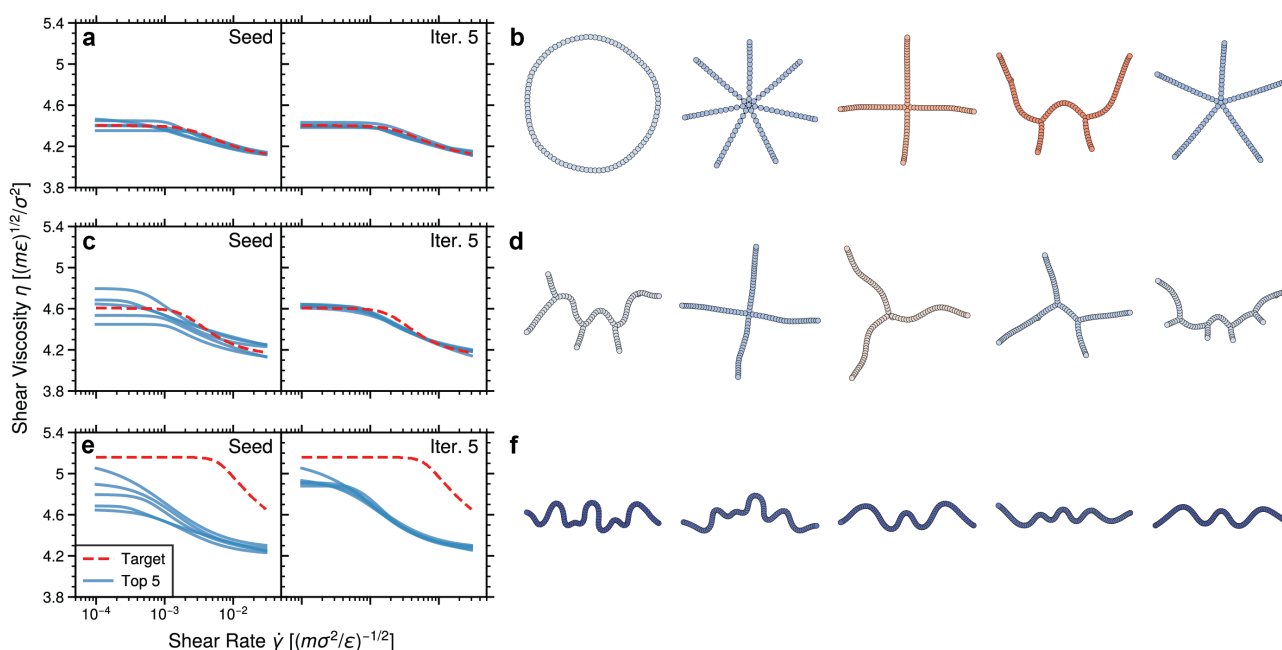


Fig. 4: Impact of active learning in targeting representative shear viscosity profiles. **a,c,e**, Target shear viscosity profiles (red dashed) compared with the five best-matching candidates before (left) and after (right) five active learning iterations for **a** low-, **c** medium-, and **e** high-difficulty targets. All profiles are fitted using the Carreau–Yasuda model. **b,d,f**, Polymer topologies of the five best candidates after active learning for the corresponding **b** low-, **d** medium-, and **f** high-difficulty targets. Node color indicates solvophobicity, ranging from $\lambda = 0$ (blue) to $\lambda = 0.5$ (red).

thinning or elevated viscosities required under the imposed material constraints (i.e., fixed molecular weights and concentrations).

We suggest that this topological convergence arises not because diverse solutions are impossible, but rather no solution can be produced from our latent space. This failure mode could provide valuable feedback for future design campaigns that rely on similar strategies. Convergence on inadequate candidates signals when constraints must be relaxed or alternative search methods should be considered. While our generative model is limited to interpolation within the training distribution, evolutionary approaches like genetic algorithms^{42,67–69} could enable less bounded searches, if anticipated and desired.

2.3 Scaling analysis enables rationale extrapolation of design space

The results of our active learning campaign indicate that high-difficulty viscosity targets likely cannot be achieved by polymers within our latent space, which is linked to particular material constraints imposed by fixed concentration and molecular weight. Analysis of unsuccessful high-difficulty targets reveals two common deficiencies: insufficient zero-shear viscosity (η_0) and premature onset of shear-thinning (Supplementary Information Figs. S12 and S13). Rather than performing costly expansion and re-optimization of the latent space, we employ scaling relationships from polymer physics to rationally adjust our design parameters.

The Carreau-Yasuda model,^{70,71} which parameterizes our viscosity targets (see also Eq. 16 and Section 4.3.2), provides guidance as to how to modify our designs. Delaying shear-thinning onset requires reducing the relaxation time τ , which scales approximately with the Zimm relaxation time τ_Z .⁷² For linear polymers in dilute solution, $\tau_Z \propto \eta_s R^3 / (k_B T) \propto N^{3\nu}$, where N is the degree of polymerization, η_s the solvent viscosity, R the coil size, and ν the Flory exponent. While altering solvent quality can affect τ through ν , reducing molecular weight provides more direct control, addressing the premature shear-thinning in our initial candidates. However, shorter chains will also tend to reduce zero-shear viscosity. This is expected by the relationship $\eta_0 \approx \eta_s(1 + c[\eta])$, where the intrinsic viscosity $[\eta] \propto N^{3\nu-1}$ for Zimm polymers. To compensate, we can increase polymer concentration while remaining below the overlap concentration c^* . Therefore, we conclude that expanding the accessible rheological regime can be achieved by tandem adjustments to molecular weight and concentration.

We implement this approach by first identifying polymers within the dataset produced from active learning whose viscosity profiles most closely approximate the challenging targets. Topology-preserving coarse-graining systematically reduces molecular weight while maintaining topological features (Fig. 5a). We then adjust polymer concentrations according to Zimm scaling to approach target rheological behavior. Specifically, we leverage the em-

pirical fit for the scaling exponent $\nu(\lambda)$ provided in Supplementary Information Section S2. Using this relationship, we simulate polymers with effective molecular weights reduced to 1/4 and 1/8 of their original values and evaluate their rheological response at concentrations of 0.05, 0.06, 0.07, 0.08, and 0.09 σ^{-3} . This strategy substantially improves correspondence with previously unattainable targets, particularly for zero-shear viscosity and shear-thinning onset (Figs. 5b-d). Remaining discrepancies in transition sharpness and power-law behavior likely require refinement of solvophobicity or topology. These results demonstrate how data-driven discovery and physics-based design principles synergistically expand the accessible polymer design space.

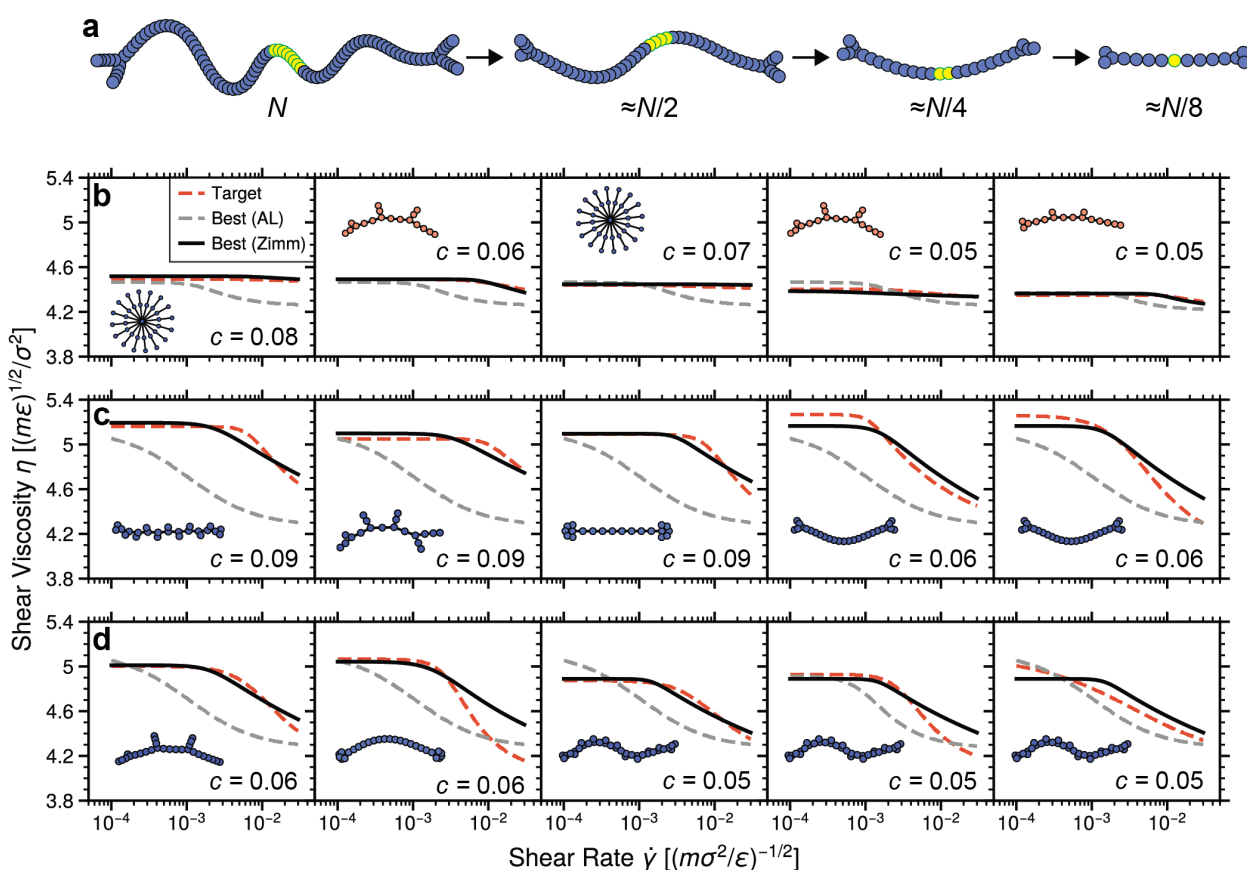


Fig. 5: Physics-based extrapolation for challenging targets. **a**, Schematic of topology-preserving coarse-graining with molecular weight approximately halved at each level. This procedure is used to produce polymers with lesser molecular weight without altering the original architectural class. **b–d**, Target viscosity profiles (red dashed) compared with results from physics-based extrapolation (black) and the best result from active learning (gray dashed), for previously unmatched **b** medium-difficulty and **c,d**, high-difficulty targets. All curves fitted using the Carreau–Yasuda model.

2.4 Conformational analysis reveals that transverse polymer dimensions govern shear viscosity

We now return to the observation that topologically and chemically distinct polymers can produce nearly identical rheological profiles (see Supplementary Information Section S10 and Figs. S8-S13). While such degeneracy might be expected given the vastness of the design space, it underscores the non-trivial nature of polymer rheology and motivates deeper investigation into the essential physical principles governing flow behavior. We therefore perform a data-driven analysis of the chain conformations under shear to identify if there are shared physical features of these polymers that explain the congruence in functional response.

Analysis of the gyration tensor reveals a striking pattern (Fig. 6a), in which, at constant shear stress of $9 \times 10^{-4} \varepsilon/\sigma^3$, shear viscosity correlates strongly with the transverse components G_{yy} and G_{zz} ($r \geq 0.95$), but only weakly with the flow-aligned component G_{xx} . The geometric mean $\sqrt{G_{yy}G_{zz}}$, representing effective cross-sectional area perpendicular to flow, shows the strongest correlation with viscosity ($r = 0.96$). In contrast, global shape descriptors like asphericity and prolateness exhibit negligible correlation, indicating that absolute transverse dimensions matter more than overall chain anisotropy. Representative simulation snapshots (Figs. 6b-e) reveal that, despite differences in topology and solvophobicity, polymers adopt comparable transverse dimensions, evident from the similar spread of the highlighted chains within the reference circles in the yz view. In contrast, their extent along the flow direction x varies markedly, reflecting the weaker correlation between G_{xx} and viscosity. A similar analysis under higher shear (Supplementary Information Fig. S6) at constant shear stress of $2 \times 10^{-3} \varepsilon/\sigma^3$ finds G_{yy} and G_{zz} still strongly correlated with viscosity ($r \geq 0.94$), while the correlation for G_{xx} drops to $r = 0.69$. The geometric mean $\sqrt{G_{yy}G_{zz}}$ again gives the strongest correlation ($r = 0.96$), confirming its role as a key descriptor.

To better elucidate these correlations and provide some sense of design principles, we consider whether key parameters from the Carreau–Yasuda model can be estimated using

only topological descriptors of the polymer and its solvophobicity. In particular, we build models using Lasso regression and then apply SHAP analysis to identify influential features. We find that η_0 depends primarily on graph diameter ϕ_{diam} and solvophobicity λ . These topological measures effectively capture spatial extent of the graph and also polymer size. Both are intuitive under zero-shear conditions. Meanwhile, the relaxation time τ shows similar dependencies, but the average node betweenness centrality C_{bet} emerges as the most influential metric. At $\lambda = 0$ and fixed molecular weights, C_{bet} correlates linearly and positively with $\langle R_g^2 \rangle$,²⁴ indicating that polymers with larger effective size tend to respond more slowly. The consistent negative influence of λ reflects how chain compaction reduces both static size and dynamic response. This analysis demonstrates how graph-theoretic descriptors can predict rheological behavior, providing a computationally efficient screening tool for polymer design.

While the importance of cross-sectional area for hydrodynamic drag appears intuitive, this relationship has not been clearly demonstrated across diverse polymer topologies. Previous studies have established that, for linear polymers in dilute solutions, the polymer contribution to shear viscosity scales with the square of the polymer thickness in the shear gradient direction (i.e., G_{zz} or δ_2^2).⁷³ Prior studies on topology effects on polymer rheology have been largely restricted to variations within single architectural classes, such as changing chain length and stiffness for linear polymers³³ or varying composition for star copolymers.³⁶ This makes it difficult to disentangle architecture-specific effects from universal principles. Our machine learning framework, by simultaneously considering linear, branched, star, and dendritic polymers, could enable analysis that would be difficult to construct through human intuition alone. The consistent dominance of transverse dimensions across all topologies establishes a unifying principle for interpreting shear viscosity.

3 Discussion

We utilized machine learning to design polymers that yield targeted shear-rate-dependent viscosity profiles. This provides a major advance for polymer informatics, as a showcase of navigating a vast design space, defined by topology and monomer solvophobicity, to achieve complex functional behavior, going beyond scalar properties of linear homopolymers. While active learning with graph-based generative modeling successfully fine-tuned polymer topologies and solvophobicity to satisfy most targets, we satisfied high-difficulty targets by using simple scaling laws to rationalize how to extrapolate from our design space. These results highlight how machine learning excels at interpolative design within established bounds, while physics-based reasoning provides essential extrapolative power. Future advances will benefit from tighter integration of these complementary strengths rather than treating them as separate tools. This is the central motivation of physics-informed machine learning,^{60,61,63,74} although such strategies still require thoughtful human construction to be successful.

We also suggest several other implications and directions for future research. First, extensions to semi-dilute or melt regimes and multi-component systems represent natural next steps. However, it is worth noting that our modeling scheme treated solvophobicity as a simple tunable parameter. Realizing similar results in real systems will require mapping specific monomer chemistries to effective interactions; this may be resolved by the use of chemically specific coarse-grained models^{75–77} or phenomenological heuristics.^{78,79} Additionally, while the architectural classes included are motivated by experimental works, synthesizing precise topologies would be challenging.⁸⁰ Future implementations can, however, incorporate specific constraints⁴³ or leverage information regarding architectural ensembles;⁸¹ this also motivates tools for reliably modeling synthetic outcomes.^{46,82–85} Nevertheless, the success here suggests similar strategies could accelerate discovery in other soft matter systems where functional behavior can be controlled by architectural design or assembly.

4 Methods

4.1 Molecular dynamics simulations

4.1.1 General simulation protocols

All simulations are conducted using HOOMD-blue.⁸⁶ All variables are expressed such that mass m , length σ , and energy ε define the unit system. The reduced time unit is given by $\tau = \sqrt{m\sigma^2/\varepsilon}$, and the equations of motion are integrated via the velocity Verlet algorithm with timestep $\Delta t_{\text{MD}} = 0.01 \tau$. To capture hydrodynamic interactions, a hybrid simulation framework is used, where molecular dynamics (MD) describes the polymer (solute) dynamics, and multiparticle collision dynamics (MPCD) captures the effect of the solvent.^{25,26,30,87} In the MPCD scheme, the solvent is modeled as a collection of point particles that evolve through alternating ballistic streaming and stochastic collision steps. During the streaming step, solvent particle positions are propagated over a time interval $\Delta t_{\text{MPCD}} = 0.1 \tau$. In the collision step, solute and solvent particles are sorted into cubic cells of edge length σ . To ensure Galilean invariance, the collision grid is shifted before each step by a random vector uniformly sampled from $[-\sigma/2, \sigma/2]^3$. Velocities are updated using an Andersen thermostat scheme.^{28,88}

4.1.2 Non-equilibrium simulations with shear flow

To calculate shear-rate-dependent viscosities, simulations are performed with steady-state shear flow. All simulations use a cubic simulation cell with edge length $L_{\text{box}} = 120 \sigma$ and periodic boundary conditions. The number density of solvent particles is set to $\rho_s = 5 \sigma^{-3}$, and the monomer density to $\rho_m = 0.02 \sigma^{-3}$, corresponding to approximately 2 wt% polymer concentration. The system is first evolved for 10^5 MD steps without MPCD solvent, followed by 5×10^5 steps with solvent present, all under quiescent conditions. Production runs span 2×10^6 steps for systems with $\langle \Delta p_x \rangle > 0.01 p_0$ (see Eq. 1), and 10^7 steps for lower shear stress conditions, to ensure statistical convergence of the velocity profile (see Supplementary

Information Section S13 for data collection details).

To achieve steady shear flow, we use the reverse perturbation method,^{89,90} which introduces a net momentum flux via velocity exchanges between solvent particles. The simulation cell is divided into slabs of width σ along the z -axis (velocity gradient direction), with flow along the x -axis. Two slabs, located at $z = 0$ and $z = L_{\text{box}}/2$, are designated for velocity exchange. In each MPCD collision step, the particle with the largest positive v_x in the first slab and the particle with the largest negative v_x in the second are identified and their velocities exchanged to generate a net momentum flux. Because solvent particles have equal mass, each exchange conserves both linear momentum and kinetic energy.

The magnitude of the imposed shear stress depends on the number and frequency of exchanges. The average shear stress is computed as

$$\langle \tau_{\text{stress}} \rangle = \frac{\langle \Delta p_x \rangle}{2\Delta t L_{\text{box}}^2}, \quad (1)$$

where $\langle \Delta p_x \rangle$ is the average x -momentum exchanged per time step Δt . The shear rate is obtained from the linear slope of the steady-state velocity profile:

$$\dot{\gamma} = \frac{dv_x}{dz}, \quad (2)$$

where v_x is the velocity component in the flow direction and z is the coordinate in the gradient direction. The applied momentum exchange $\langle \Delta p_x \rangle$ is selected from the set $[0.004, 0.006, 0.008, 0.01, 0.03, 0.06, 0.12, 0.18, 0.24, 0.3, 0.4, 0.6, 0.8, 1] \times p_0$, where $p_0 = L_{\text{box}}^2 \times 10^{-4}$.

4.1.3 Polymer model

Each polymer consists of $N \leq 100$ beads, where N denotes the degree of polymerization. Each bead has diameter σ and mass $M = 5m$. The total potential energy U of a polymer

configuration $\mathbf{r}^N \equiv \{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ is given by

$$U(\mathbf{r}^N) = \sum_{i,j \text{ bonded}} U_{\text{vib}}(r_{ij}) + \sum_{i < j} U_{\text{nb}}(r_{ij}), \quad (3)$$

where $r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$.

Nonbonded interactions are modeled using the Ashbaugh–Hatch potential:⁹¹

$$U_{\text{nb}}(r_{ij}) = \begin{cases} U_{\text{LJ}}(r_{ij}) + (1 - \lambda_{ij})\varepsilon_{ij}, & \text{if } r_{ij} \leq 2^{1/6}\sigma, \\ \lambda_{ij}U_{\text{LJ}}(r_{ij}), & \text{otherwise.} \end{cases} \quad (4)$$

Here, U_{LJ} is the Lennard-Jones (LJ) potential,

$$U_{\text{LJ}}(r_{ij}) = 4\varepsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right], \quad (5)$$

where $\sigma_{ij} = \sigma$ and $\varepsilon_{ij} = \varepsilon$. The parameter λ_{ij} modulates solvophobicity and is assigned a uniform value λ for each polymer. When $\lambda = 1$, the interaction recovers the full LJ potential; when $\lambda = 0$, it becomes purely repulsive. To remain in the good-solvent regime, we restrict λ to $[0, 0.5]$, based on R_g scaling analysis for linear polymers (Supplementary Information Section S2).⁶ Bonded interactions are described by the finitely extensible nonlinear elastic (FENE) potential:⁹²

$$U_{\text{vib}}(r_{ij}) = \begin{cases} -\frac{1}{2}K_{ij} \left(R_{ij}^{(0)} \right)^2 \ln \left[1 - \left(\frac{r_{ij}}{R_{ij}^{(0)}} \right)^2 \right], & r_{ij} \leq R_{ij}^{(0)} \\ \infty, & r_{ij} > R_{ij}^{(0)}, \end{cases} \quad (6)$$

where $K_{ij} = 30\varepsilon/\sigma^2$ is the spring constant, and $R_{ij}^{(0)} = 1.5\sigma$ is the maximum bond extension.

4.2 Details of Latent-space Representation

4.2.1 Description of dataset

We augmented our previously developed `TopoRg-1342` dataset to incorporate solvophobicity as an additional design parameter.²⁴ The original dataset contains mean squared radius of gyration $\langle R_g^2 \rangle$ data for 1,342 polymers spanning six architectural classes: 11 each for linear and cyclic, and 330 each for $\alpha\omega$ -branched, comb, star, and dendrimer topologies. All polymers have degrees of polymerization between 90 and 100. In the original work, solvophobicity was fixed at $\lambda = 0$, representing purely repulsive interactions. To explore rheological behavior across varying solvent conditions, we assigned each polymer topology five solvophobicity values ($\lambda \in [0, 0.5]$). This range was chosen to maintain good-solvent conditions at low polymer concentrations while introducing sufficient variation in chain compaction. The augmentation yields 6,710 unique polymer-solvophobicity combinations.

To help disambiguate topological effects from compaction induced by solvophobicity, we also calculate additional properties for the 6,710 systems. In particular, in addition to $\langle R_g^2 \rangle$, which was featured in the prior dataset, we add N , λ , and the chain virial number Ξ . The latter two properties report on intermolecular interactions. These properties are used to guide the organization of the universal latent space representation of polymers that captures both topological and solvophobicity variations. Among the properties used here, Ξ is one newly introduced in this work, and we describe it subsequently.

We define the chain virial number as

$$\Xi \equiv \frac{\hat{B}_2^{cc}}{V_c} \approx \frac{N^2 \hat{B}_2^{bb}}{N\sigma^3} = \frac{N \hat{B}_2^{bb}}{\sigma^3}, \quad (7)$$

where $\hat{B}_2^{cc}$ and $\hat{B}_2^{bb}$ denote approximate chain–chain and bead–bead second virial coefficients, respectively. The bead–bead term is computed as

$$\hat{B}_2^{bb} = 2\pi \int_0^\infty [1 - e^{-\beta U(r)}] r^2 dr, \quad (8)$$

with $\beta = 1/(k_{\text{B}}T)$ and $U(r)$ the pair potential. In Eq. (7), the denominator is a reference contour volume per chain, $V_{\text{c}} \approx N\sigma^3$. This definition makes Ξ dimensionless. For interpretability, both numerator and denominator can be normalized by the pervaded volume R_{g}^3 , $\Xi = \frac{\hat{B}_2^{\text{cc}}/R_{\text{g}}^3}{V_{\text{c}}/R_{\text{g}}^3} = \frac{\hat{A}_2}{\phi_{\text{c}}}$, where $\hat{A}_2$ is a reduced chain-level virial coefficient and $\phi_{\text{c}} \equiv V_{\text{c}}/R_{\text{g}}^3$ is the contour-volume fraction within the pervaded volume. The quantity ϕ_{c} scales with the overlap concentration, since $c^* \propto 1/R_{\text{g}}^3$. Thus, the chain virial number balances drivers for inter-chain closeness, with the numerator capturing interaction strength and the denominator reflecting intrinsic chain size. For $\Xi = 0$, chains are simply as far apart as expected from their concentration. For $\Xi < 0$, chains are closer due to net attraction, and for $\Xi > 0$ they are farther due to net repulsion. We suspect that many alternatives to Ξ would be similarly effective/informative.

4.2.2 Data preprocessing

Polymers are represented as graphs $\mathcal{G} = (V, E)$, where V denotes nodes and E denotes edges, following our previous study.²⁴ To standardize graph sizes, we introduce “ghost” nodes with zero-edge connections, ensuring all graphs have 100 nodes, as polymer bead count range from 90 to 100. This padding approach aligns with prior methods.^{93,94} Each polymer bead is described by an adjacency vector $\mathbf{a}_i \in \mathbb{R}^{100}$, where $\mathbf{a}_i = 1$ if node i is connected to the current node and $\mathbf{a}_i = 0$ otherwise. Since all bonds are equivalent, edge features are excluded. Additionally, polymers are characterized by a 12-dimensional topological descriptor vector $\mathbf{t} \in \mathbb{R}^{12}$, obtained by extending the previously defined 11-dimensional representation²⁴ with the inclusion of λ to capture solvophobicity. The original 11 descriptors comprise the number of nodes N , number of edges N_e , average node degree $\bar{d}$, average neighbor degree $\bar{d}_{\text{nb}}$, graph density δ , graph diameter ϕ_{diam} , graph radius ϕ_{rad} , algebraic connectivity λ_{alg} , average degree centrality C_{deg} , average betweenness centrality C_{bet} , and degree assortativity r_{deg} . For polymer reconstruction, each polymer is assigned an adjacency matrix $\mathbf{A} \in \mathbb{R}^{100 \times 100}$, where $A_{ij} = 1$ if nodes i and j are connected and $A_{ij} = 0$ otherwise. Additionally, we introduce two auxiliary tasks: a regression task with continuous labels

$\mathbf{y}_r \in \mathbb{R}^4$ for $\langle R_g^2 \rangle$, λ , N , and Ξ , and a classification task with a one-hot encoded architectural label $\mathbf{y}_t \in \mathbb{R}^6$.

For the purpose of training the variational autoencoder (VAE, see also Section 4.2.3), the total dataset of 6,710 polymers is partitioned into training (4,294, 64%), validation (1,074, 16%), and testing (1,342, 20%) subsets. The generation of subsets uses a stratified splitting to ensure proportional representation of all polymer architectural classes. To prevent data leakage, polymers with the same topology but different solvophobicity values are assigned to the same subset.

4.2.3 VAE model architectures

As a starting point, we adopt the architecture of the best-performing model from our previous study.²⁴ The VAE architecture for TopoGNN is illustrated in Fig. S2. The encoder maps the input data into a latent space representation by processing graph inputs, represented by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{100 \times 100}$ and a node feature matrix $\mathbf{X} \in \mathbb{R}^{100 \times 100}$, using a Graph Isomorphism Network (GIN) encoder.⁹⁵ With its two graph convolutional layers, the GIN encoder produces a 32-dimensional feature vector $\mathbf{h}_g$. A dense neural network similarly converts the topological descriptor vector into a 32-dimensional feature vector $\mathbf{h}_t$. These feature vectors are concatenated to form a combined vector $\mathbf{h} \in \mathbb{R}^{64}$, which is further processed by dense layers to generate the parameters, the mean $\boldsymbol{\mu}$ and the log variance $\log \boldsymbol{\sigma}^2$, of the latent Gaussian distribution. The latent space embedding $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma})$ is 8-dimensional. Finally, the decoder samples from $\mathbf{z}$ to reconstruct the data: a convolutional neural network reconstructs the adjacency matrix $\hat{\mathbf{A}}$, and two distinct neural networks predict $\hat{\mathbf{y}}_r$ and $\hat{\mathbf{y}}_t$.

4.2.4 Loss functions

Training of the VAE uses a composite loss function $\mathcal{L}_{\text{VAE}}$

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{Rec}} + \mathcal{L}_{\text{KL}} + \lambda_{\text{Reg}} \mathcal{L}_{\text{Reg}} + \lambda_{\text{Cls}} \mathcal{L}_{\text{Cls}}, \quad (9)$$

which features terms associated with reconstruction, $\mathcal{L}_{\text{Rec}}$ via binary cross-entropy (BCE); Kullback-Leibler (KL) divergence, $\mathcal{L}_{\text{KL}}$; regression for y_r , $\mathcal{L}_{\text{Reg}}$; and classification for y_t via cross-entropy (CE), $\mathcal{L}_{\text{Cls}}$. In Eq. (9), λ_{Reg} and λ_{Cls} are hyperparameter weights that are adjustable to modulate the contributions of regression and classification tasks, respectively.

The individual loss terms are defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{Rec}} &= \text{BCE}(\mathbf{A}, \hat{\mathbf{A}}) \\ &= - \sum_{i=1}^{100} \sum_{j=1}^{100} A_{ij} \log(\hat{A}_{ij}) + (1 - A_{ij}) \log(1 - \hat{A}_{ij}),\end{aligned}\tag{10}$$

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= D_{\text{KL}}(\mathbf{z} || \mathcal{N}(0, \mathbf{I})) \\ &= -\frac{1}{2} \sum_{i=1}^8 (1 + \log(\sigma_i^2) - \sigma_i^2 - \mu_i^2),\end{aligned}\tag{11}$$

$$\begin{aligned}\mathcal{L}_{\text{Reg}} &= \text{MAE}(\mathbf{y}_r, \hat{\mathbf{y}}_r) \\ &= |\mathbf{y}_r - \hat{\mathbf{y}}_r|,\end{aligned}\tag{12}$$

$$\begin{aligned}\mathcal{L}_{\text{Cls}} &= \text{CE}(\mathbf{y}_t, \hat{\mathbf{y}}_t) \\ &= - \sum_{i=1}^6 y_{t,i} \log(\hat{y}_{t,i}).\end{aligned}\tag{13}$$

4.2.5 Model training and hyperparameter tuning

Model training is performed using TensorFlow⁹⁶ and the Adam optimizer⁹⁷ for 1000 epochs. A grid search over 500 hyperparameter configurations is conducted. The search space includes batch sizes $\{32, 64, 128, 256\}$, learning rates $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$, and regularization weights $\lambda_{\text{Reg}}, \lambda_{\text{Cls}} \in \{0.01, 0.1, 1, 10, 100\}$.

To guide model selection, we adopt a comprehensive evaluation score (CES), following our prior work,²⁴ to quantify overall performance across reconstruction, regression, and

classification tasks:

$$\text{CES} \equiv \sqrt{\overline{(1 - \text{BACC})^2} + \overline{(\text{KL})^2} + \overline{(1 - R^2)^2} + \overline{(1 - F_1)^2}}, \quad (14)$$

where $\bar{a}$ denotes min-max normalization of metric a across all evaluated configurations. Here, BACC measures the balanced accuracy of binarized adjacency matrix reconstruction (threshold = 0.5), R^2 denotes the average coefficient of determination for $\langle R_g^2 \rangle$, λ , N , and Ξ , and F_1 accounts for class imbalance in topology classification. KL divergence serves as a proxy for latent space regularity. The CES thus provides a summary of task-level performance, and the configuration yielding the lowest CES is selected for subsequent analysis.

4.3 Active learning discovery

We implement an active learning (AL) framework to identify polymer candidates with prescribed shear-rate-dependent viscosity profiles. The iterative workflow comprises four components: (1) a Gaussian process (GP) surrogate that predicts viscosity profiles from latent polymer representations; (2) a Bayesian optimization (BO) scheme to select informative latent vectors; (3) a variational autoencoder (VAE) decoder that maps latent vectors to polymer topologies and solvophobicity parameters as prospective candidates; and (4) non-equilibrium MPCD simulations to characterize the rheological response of selected candidates. After curation of an initial seed dataset, five iterations of active learning are conducted, with each iteration defined by the characterization of 30 unique polymers (one per target); this newly simulated data is used to retrain the surrogate model to begin a new iteration of acquisition. The following sections provide more details regarding key components.

4.3.1 Seed dataset selection

Seed points are selected from the polymer latent space using a maximin diversity criterion,^{98,99} designed to ensure broad coverage. The latent vectors $\mathbf{Z} \in \mathbb{R}^{N \times 8}$ are obtained from a trained VAE and normalized to the unit hypercube $[0, 1]^8$ using min-max scaling. Starting

from a random seed $\mathbf{s}_1$, subsequent points are chosen greedily to maximize their minimum Euclidean distance from the existing set:

$$\mathbf{s}_k = \arg \max_{i \in \mathcal{I} \setminus \{\mathbf{s}_1, \dots, \mathbf{s}_{k-1}\}} \min_{j \in \{\mathbf{s}_1, \dots, \mathbf{s}_{k-1}\}} d(\mathbf{z}_i, \mathbf{z}_j), \quad (15)$$

where $\mathcal{I} = \{1, \dots, N\}$ is the index set and $d(\cdot, \cdot)$ denotes Euclidean distance. This procedure continues until 30 points are selected.

4.3.2 Target viscosity profiles

To explore a controlled and representative spectrum of shear-thinning rheology, we generate synthetic viscosity targets using a model originally developed by Carreau and extended by Yasuda:^{70,71}

$$\eta(\dot{\gamma}) = \eta_\infty + (\eta_0 - \eta_\infty) [1 + (\tau \dot{\gamma})^a]^{\frac{n-1}{a}}, \quad (16)$$

where η_0 and η_∞ are the zero- and infinite-shear viscosities, respectively; τ is the relaxation time marking the onset of shear-thinning; a controls the breadth of the transition (transition sharpness); and n quantifies the degree of shear-thinning in the high-shear regime (shear-thinning exponent).

A total of 30 synthetic targets are constructed by sampling the Carreau–Yasuda model over representative parameter intervals: $\eta_0 \in [4.0, 5.3]$, $\eta_\infty \in [4.0, 4.5]$, $\tau \in [10^2, 10^3]$, $a \in [0.1, 5.0]$, and $n \in [0.01, 0.99]$. To ensure consistency with shear-thinning rheology, all target curves satisfy $\eta_0 > \eta_\infty$. Among them, a subset of 10 targets is further constrained by $\eta_0 - \eta_\infty < 0.1$ in order to approximate weak shear-thinning or near Newtonian conditions. The complete set of parameters is provided in the Supplementary Information Section S4. To enable consistent rheological analysis, all rheological simulation results are fit to Eq. (16) via nonlinear least-squares optimization implemented in SciPy.¹⁰⁰

4.3.3 Gaussian process surrogate model

We employ Gaussian process (GP) regression¹⁰¹ to map polymer latent representations ($\mathbf{z} \in \mathbb{R}^8$) to their simulated shear-rate-dependent viscosity profiles. The model predicts $\mathbf{y} \in \mathbb{R}^{100}$, representing viscosity values at 100 logarithmically spaced points across $\dot{\gamma} \in [10^{-4}, 0.03]$.

We adopt an independent multi-output GP formulation, such that each output dimension is modeled with a separate GP:

$$y_i(\mathbf{z}) \sim \mathcal{GP}(m_i(\mathbf{z}), k_i(\mathbf{z}, \mathbf{z}')), \quad i = 1, \dots, 100. \quad (17)$$

We use the Matérn kernel for each k_i due to its tunable smoothness:

$$k(\mathbf{z}_m, \mathbf{z}_n) = \frac{1}{\Gamma(\nu)2^{\nu-1}} \left(\frac{\sqrt{2\nu}}{l} d(\mathbf{z}_m, \mathbf{z}_n) \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}}{l} d(\mathbf{z}_m, \mathbf{z}_n) \right), \quad (18)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance, $K_\nu(\cdot)$ is the modified Bessel function of the second kind, and $\Gamma(\cdot)$ is the gamma function. The smoothness parameter ν is selected from $\{1.0, 1.1, \dots, 2.5\}$ by minimizing the average mean squared error (MSE) across output dimensions using three-fold cross-validation. The kernel length scale l is fixed to 1 throughout. After hyperparameter selection, the final model is retrained on the full dataset. Given a new latent vector $\mathbf{z}_{\text{new}}$, the GP yields the predicted mean and variance for each output dimension:

$$\mu_i(\mathbf{z}_{\text{new}}) = \mathbf{k}_{\text{new}}^\top (\mathbf{K}_i + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_i, \quad (19)$$

$$\sigma_i^2(\mathbf{z}_{\text{new}}) = k_i(\mathbf{z}_{\text{new}}, \mathbf{z}_{\text{new}}) - \mathbf{k}_{\text{new}}^\top (\mathbf{K}_i + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{\text{new}}, \quad (20)$$

where $\mathbf{k}_{\text{new}} \in \mathbb{R}^N$ is the kernel vector between $\mathbf{z}_{\text{new}}$ and the training points, and $\mathbf{K}_i \in \mathbb{R}^{N \times N}$ is the kernel matrix for output dimension i . The noise term σ^2 is fixed to 10^{-10} for numerical stability.

4.3.4 Bayesian optimization

Selecting polymer candidates for data acquisition is guided by Bayesian optimization (BO) using the Expected Improvement (EI) acquisition function.¹⁰² Unlike conventional EI, which targets scalar model outputs, our objective is to minimize the MSE between the predicted and target shear-rate-dependent viscosity curve. To accommodate this, we employ a target-aware variant of EI that prioritizes candidates expected to reduce discrepancy.¹⁰³ In particular, for a latent vector $\mathbf{z}$, the Gaussian process model $\mathcal{GP}$ returns a predictive mean vector $\boldsymbol{\mu}(\mathbf{z}) = [\mu_1(\mathbf{z}), \dots, \mu_d(\mathbf{z})]$ and standard-deviation vector $\boldsymbol{\sigma}(\mathbf{z}) = [\sigma_1(\mathbf{z}), \dots, \sigma_d(\mathbf{z})]$, where d denotes the number of degrees of freedom. Further, let $\Delta \mid \mathbf{z} = \|\mathcal{GP}(\mathbf{z}) - \mathbf{y}^{\text{target}}\|_2^2$ denote the squared L_2 discrepancy, which we seek to minimize in expectation. As shown by Uhrenholt et al.,¹⁰³ the random variable $\Delta \mid \mathbf{z}$ follows a noncentral chi-squared distribution and noncentrality parameter β .

$$p(\Delta \mid \mathbf{z}, \mathcal{D}) = \chi_d^2(\Delta \gamma^{-2} \mid d, \beta) \gamma^{-2}, \quad (21)$$

$$\gamma^2 = \frac{1}{d} \sum_{i=1}^d \sigma_i^2(\mathbf{z}), \quad (22)$$

$$\beta = \gamma^{-2} \sum_{i=1}^d [\mu_i(\mathbf{z}) - y_i^{\text{target}}]^2, \quad (23)$$

If $F_{d,\beta}(t)$ denote the cumulative distribution function of $\chi_d^2(\cdot; \beta)$ and $t = \Delta_{\min} \gamma^{-2}$, where $\Delta_{\min} = \min_{\mathbf{z}' \in \mathcal{D}} \Delta(\mathbf{z}')$, then the target-aware EI is given by

$$\mathcal{EI}(\mathbf{z}; \mathcal{D}) = \gamma^2 [t F_{d,\beta}(t) - d F_{d+2,\beta}(t) - \beta F_{d+4,\beta}(t)]. \quad (24)$$

We optimize the acquisition function using the L-BFGS-B algorithm,¹⁰⁴ as implemented in `SciPy`, with multiple random restarts to mitigate local optima. For each iteration of active learning, 30 distinct optimizations of Eq. (24) are performed by changing $\mathbf{y}^{\text{target}}$ to correspond to each of the target viscosity profiles. This produces a batch of 30 latent vectors per

iteration that are decoded to explicit polymer topology/solvophobicity combinations (Section 4.3.5). After decoding, characterization with non-equilibrium molecular simulations is performed to obtain “ground-truth” shear-rate-dependent viscosity profiles.

4.3.5 Polymer graph generation

Each latent vector $\mathbf{z}$ proposed by Bayesian optimization is decoded by the VAE into a polymer graph $\mathcal{G}$ and its associated solvophobicity parameter λ . A graph refinement procedure, adapted from our previous work,²⁴ is applied to remove topological artifacts such as isolated nodes or nonphysical ring structures, yielding a cleaned graph $\mathcal{G}_{\text{clean}}$. Since this step may alter the underlying latent representation, $\mathcal{G}_{\text{clean}}$ is subsequently re-encoded by the VAE to obtain an updated latent vector $\mathbf{z}_{\text{clean}}$. To preserve consistency with the original optimization objective, we perform re-optimization of the EI acquisition function over 10,000 random initializations, selecting the $\mathbf{z}$ that yields the smallest latent-space discrepancy $\|\mathbf{z}_{\text{clean}} - \mathbf{z}\|_1$. The corresponding polymer graph is then retained as the final candidate for simulation.

4.4 Analysis

4.4.1 Search performance evaluation

To evaluate search performance over active learning iterations, we define a normalized deviation metric that quantifies how closely the top-performing discovered candidates match the target viscosity profiles. For a given target viscosity curve $\boldsymbol{\eta}_{t,k}$ and iteration j , we collect all candidate curves identified up to and including iteration j , denoted as $\{\boldsymbol{\eta}_{c,l}\}_{l=1}^{L_j}$, where L_j is the cumulative number of discovered candidates (including those in the initial seed dataset).

We then compute the MSE between the target curve and its i best-matching candidate

curves under the squared L_2 norm:

$$d_{i,k}^{(j)} = \frac{1}{i} \sum_{\ell=1}^i \|\boldsymbol{\eta}_{t,k} - \boldsymbol{\eta}_{c,\pi_\ell}\|_2^2, \quad (25)$$

where $\pi_1, \dots, \pi_i$ index the i candidate curves with the smallest squared L_2 distances to target k among all candidates available by iteration j .

To report aggregate performance, we average this deviation across all targets k within a given difficulty group:

$$\bar{d}_i^{(j)} = \frac{1}{K} \sum_{k=1}^K d_{i,k}^{(j)}, \quad (26)$$

where K is the number of targets in the group. This metric $\bar{d}_i^{(j)}$ reflects the average rheological closeness between each target and its top- i discovered candidates at iteration j .

4.4.2 Conformational descriptors

To characterize polymer conformations under shear, we compute the gyration tensor $\mathbf{G}$ and associated scalar shape descriptors—asphericity, prolateness, and acylindricity—for each chain.¹⁰⁵ The gyration tensor of each polymer $\mathbf{G}$ is defined as

$$G_{\alpha\beta} = \frac{1}{N} \sum_{i=1}^N (r_{i,\alpha} - \bar{r}_\alpha)(r_{i,\beta} - \bar{r}_\beta), \quad (27)$$

where $r_{i,\alpha}$ is the α -component of the position of monomer i and $\bar{r}_\alpha$ is the corresponding center-of-mass coordinate. Shape descriptors are computed from the eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ of $\mathbf{G}$ using the following definitions: asphericity $b = 1 - 3I_2/I_1^2$, prolateness $S = (1/I_1^3)(3\lambda_1 - I_1)(3\lambda_2 - I_1)(3\lambda_3 - I_1)$, and acylindricity $c = (\lambda_2 - \lambda_3)/I_1$, where $I_1 = \lambda_1 + \lambda_2 + \lambda_3$ and $I_2 = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1$. In cases where $I_1 = 0$, all shape descriptors are set to zero. All quantities are evaluated under steady-state shear. Conformational data are averaged over all chains and across 500 simulation time steps, with each step corre-

sponding to 10τ . This analysis is performed over 150 polymer candidates proposed by the active learning framework. Conformational analyses are conducted under two fixed shear stresses, 9×10^{-4} and $2 \times 10^{-3} \varepsilon/\sigma^3$, corresponding to shear rates of 4.2×10^{-3} and $9.4 \times 10^{-3} (m\sigma^2/\varepsilon)^{-1/2}$, respectively.

4.4.3 Feature importance assessment

We perform feature attribution analysis using SHAP (SHapley Additive exPlanations) to identify molecular descriptors most predictive of rheological properties.¹⁰⁶ The analysis is conducted on 180 polymer candidates, comprising both the initial 30-seed dataset and those proposed by the active learning framework. For each polymer, we compute an augmented set of topological descriptors, including solvophobicity λ , and fit their shear viscosity profiles using the Carreau–Yasuda model to extract rheological parameters. We then train separate Lasso regression models (penalty coefficient $\alpha = 0.01$) to predict each rheological parameter using a normalized feature set. All features are scaled to the range $[0, 1]$ using min–max normalization prior to model training. Only features with nonzero Lasso coefficients are retained for interpretation by SHAP values, which indicate whether a given feature contributed positively or negatively to a given prediction.

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Data Availability

The data associated with this study are publicly available at [DOI:10.5281/zenodo.16696725](https://doi.org/10.5281/zenodo.16696725).

Code Availability

The code used in this study is publicly available at <https://github.com/webbtheosim/cg-topo-solv>.

Supplementary Information

MPCD solvent viscosity; solvophobicity-dependent polymer scaling; generative model architecture and performance; Carreau–Yasuda viscosity fitting; space-filling and active learning comparisons; SHAP analysis; chain conformation under shear; top candidate polymers and viscosity curves.

References

- (1) Sorbie, K. S. *Polymer-improved oil recovery*; Springer Science & Business Media, 2013.
- (2) Gbadamosi, A.; Patil, S.; Kamal, M. S.; Adewunmi, A. A.; Yusuff, A. S.; Agi, A.; Oseh, J. Application of Polymers for Chemical Enhanced Oil Recovery: A Review. *Polymers* **2022**, *14*, 1433, DOI: 10.3390/polym14071433.
- (3) Firozjahi, A. M.; Saghafi, H. R. Review on chemical enhanced oil recovery using polymer flooding: Fundamentals, experimental and numerical simulation. *Petroleum* **2020**, *6*, 115–122, DOI: 10.1016/j.petlm.2019.09.003.
- (4) Saha, D.; Bhattacharya, S. Hydrocolloids as thickening and gelling agents in food: a critical review. *Journal of food science and technology* **2010**, *47*, 587–97, DOI: 10.1007/s13197-010-0162-6.

-
- (5) Dickinson, E. Hydrocolloids at interfaces and the influence on the properties of dispersed systems. *Food Hydrocolloids* **2003**, 17, 25–39, DOI: 10.1016/s0268-005x(01)00120-5.
- (6) Rubinstein, M.; Colby, R. H. *Polymer Physics*; Oxford University Press, 2003; DOI: 10.1093/oso/9780198520597.001.0001.
- (7) Ferry, J. D. *Viscoelastic Properties of Polymers*, 3rd ed.; Wiley, 1980.
- (8) Hieber, C. A.; Chiang, H. H. Shear-rate-dependence modeling of polymer melt viscosity. *Polymer Engineering & Science* **1992**, 32, 931–938, DOI: 10.1002/pen.760321404.
- (9) Soong, D.; Shen, M. Shear-rate-dependent viscosity of non-Newtonian suspensions and entangled polymer systems. *Polymer Engineering & Science* **1980**, 20, 1177–1180, DOI: 10.1002/pen.760201711.
- (10) Kulicke, W. M.; Kniewske, R. The shear viscosity dependence on concentration, molecular weight, and shear rate of polystyrene solutions. *Rheologica Acta* **1984**, 23, 75–83, DOI: 10.1007/bf01333878.
- (11) Cui, T.; Liu, S.; Huang, Q. Nonlinear shear and extensional rheology of entangled polystyrene solutions with linear and star styrene oligomeric solvents. *Rheologica Acta* **2025**, 1–9, DOI: 10.1007/s00397-025-01483-1.
- (12) Steller, R.; Iwko, J. New generalized Newtonian fluid models for quantitative description of complex viscous behavior in shear flows. *Polymer Engineering & Science* **2018**, 58, 1446–1455, DOI: 10.1002/pen.24741.
- (13) Chitrangad, B.; Middleman, S. Shear rate dependent viscosities of PPO-PS polymer blend-solvent systems. *Macromolecules* **1981**, 14, 352–354, DOI: 10.1021/ma50003a024.
- (14) Kalashnikov, V. N. Shear-rate dependent viscosity of dilute polymer solutions. *Journal of Rheology* **1994**, 38, 1385–1403, DOI: 10.1122/1.550550.

-
- (15) Mucha, M. Rheological characteristics of semi-dilute chitosan solutions. *Macromolecular Chemistry and Physics* **1997**, 198, 471–484, DOI: 10.1002/macp.1997.021980220.
- (16) Jackson, G. T.; Balhoff, M. T.; Huh, C.; Delshad, M. CFD-based representation of non-Newtonian polymer injectivity for a horizontal well with coupled formation-wellbore hydraulics. *Journal of Petroleum Science and Engineering* **2011**, 78, 86–95, DOI: 10.1016/j.petrol.2011.05.007.
- (17) Bird, R.; Giacomini, A. Polymer Fluid Dynamics: Continuum and Molecular Approaches. *Annual Review of Chemical and Biomolecular Engineering* **2016**, 7, 479–507, DOI: 10.1146/annurev-chembioeng-080615-034536.
- (18) Sefiddashti, M. H. N.; Edwards, B. J.; Khomami, B. Elucidating the Molecular Rheology of Entangled Polymeric Fluids via Comparison of Atomistic Simulations and Model Predictions. *Macromolecules* **2019**, 52, 8124–8143, DOI: 10.1021/acs.macromol.9b01099.
- (19) Behbahani, A. F.; Schneider, L.; Rissanou, A.; Chazirakis, A.; Bačová, P.; Jana, P. K.; Li, W.; Doxastakis, M.; Políńska, P.; Burkhart, C.; Müller, M.; Harmandaris, V. A. Dynamics and Rheology of Polymer Melts via Hierarchical Atomistic, Coarse-Grained, and Slip-Spring Simulations. *Macromolecules* **2021**, 54, 2740–2762, DOI: 10.1021/acs.macromol.0c02583.
- (20) Edwards, B. J.; Sefiddashti, M. H. N.; Khomami, B. Atomistic simulation of shear flow of linear alkane and polyethylene liquids: A 50-year retrospective. *Journal of Rheology* **2022**, 66, 415–489, DOI: 10.1122/8.0000365.
- (21) Doi, M.; Edwards, S. F. *The Theory of Polymer Dynamics*; International Series of Monographs on Physics; Oxford University Press: Oxford, 1988; Vol. 73.

-
- (22) Todd, B. D.; Daivis, P. J. *Nonequilibrium Molecular Dynamics: Theory, Algorithms and Applications*; Cambridge University Press: Cambridge, 2017.
- (23) Liang, H.; Webb, M. A.; Chawathe, M.; Bendejacq, D.; Pablo, J. J. d. Understanding the Structure and Rheology of Galactomannan Solutions with Coarse-Grained Modeling. *Macromolecules* **2023**, *56*, 177–187, DOI: 10.1021/acs.macromol.2c01781.
- (24) Jiang, S.; Dieng, A. B.; Webb, M. A. Property-guided generation of complex polymer topologies using variational autoencoders. *npj Computational Materials* **2024**, *10*, 139, DOI: 10.1038/s41524-024-01328-0.
- (25) Malevanets, A.; Kapral, R. Mesoscopic model for solvent dynamics. *The Journal of Chemical Physics* **1999**, *110*, 8605–8613, DOI: 10.1063/1.478857.
- (26) Malevanets, A.; Kapral, R. Solute molecular dynamics in a mesoscale solvent. *The Journal of Chemical Physics* **2000**, *112*, 7260–7269, DOI: 10.1063/1.481289.
- (27) Ihle, T.; Kroll, D. M. Stochastic rotation dynamics: A Galilean-invariant mesoscopic model for fluid flow. *Physical Review E* **2001**, *63*, 020201, DOI: 10.1103/physreve.63.020201.
- (28) Allahyarov, E.; Gompper, G. Mesoscopic solvent simulations: Multiparticle-collision dynamics of three-dimensional flows. *Physical Review E* **2002**, *66*, 036702, DOI: 10.1103/physreve.66.036702.
- (29) Mussawisade, K.; Ripoll, M.; Winkler, R. G.; Gompper, G. Dynamics of polymers in a particle-based mesoscopic solvent. *The Journal of Chemical Physics* **2005**, *123*, 144905, DOI: 10.1063/1.2041527.
- (30) Gompper, G.; Ihle, T.; Kroll, D. M.; Winkler, R. G. In *Advanced Computer Simulation Approaches for Soft Matter Sciences III*; Holm, C., Kremer, K., Eds.; Advances in Polymer Science; Springer, Berlin, Heidelberg, 2009; Vol. 221; pp 1–87, DOI: 10.1007/978-3-540-87706-6_1.

-
- (31) Pattnayak, P. K.; Kumar, A.; Tomar, G. Diffusion Dynamics of Star-Shaped Macromolecules in Dilute Solutions. *Macromolecules* **2024**, *57*, 6657–6665, DOI: 10.1021/acs.macromol.4c00407.
- (32) Huang, C.-C.; Winkler, R. G.; Sutmann, G.; Gompper, G. Semidilute Polymer Solutions at Equilibrium and under Shear Flow. *Macromolecules* **2010**, *43*, 10107–10116, DOI: 10.1021/ma101836x.
- (33) Nikoubashman, A.; Howard, M. P. Equilibrium Dynamics and Shear Rheology of Semiflexible Polymers in Solution. *Macromolecules* **2017**, *50*, 8279–8289, DOI: 10.1021/acs.macromol.7b01876.
- (34) Chen, R.; Kotkar, S. B.; Poling-Skutvik, R.; Howard, M. P.; Nikoubashman, A.; Conrad, J. C.; Palmer, J. C. Nanoparticle dynamics in semidilute polymer solutions: Rings versus linear chains. *Journal of Rheology* **2021**, *65*, 745–755, DOI: 10.1122/8.0000223.
- (35) Jaramillo-Cano, D.; Formanek, M.; Likos, C. N.; Camargo, M. Star Block-Copolymers in Shear Flow. *The journal of physical chemistry. B* **2018**, *122*, 4149–4158, DOI: 10.1021/acs.jpcb.7b12229.
- (36) Jaramillo-Cano, D.; Camargo, M.; Likos, C. N.; Gârlea, I. C. Dynamical Properties of Concentrated Suspensions of Block Copolymer Stars in Shear Flow. *Macromolecules* **2020**, *53*, DOI: 10.1021/acs.macromol.0c01365.
- (37) Ramesh, P. S.; Patra, T. K. Polymer sequence design via molecular simulation-based active learning. *Soft Matter* **2022**, *19*, 282–294, DOI: 10.1039/d2sm01193j.
- (38) Carrillo, J.-M. Y.; Parambil, V.; Patra, T. K.; Chen, Z.; Russell, T. P.; Sankaranarayanan, S. K. R. S.; Sumpter, B. G.; Batra, R. Accelerated Sequence Design of Star Block Copolymers: An Unbiased Exploration Strategy via Fusion of Molecular Dy-

-
- namics Simulations and Machine Learning. *The journal of physical chemistry. B* **2024**, *128*, 4220–4230, DOI: 10.1021/acs.jpcc.3c08110.
- (39) Shmilovich, K.; Mansbach, R. A.; Sidky, H.; Dunne, O. E.; Panda, S. S.; Tovar, J. D.; Ferguson, A. L. Discovery of Self-Assembling π -Conjugated Peptides by Active Learning-Directed Coarse-Grained Molecular Simulation. *The Journal of Physical Chemistry B* **2020**, *124*, 3873–3891, DOI: 10.1021/acs.jpcc.0c00708.
- (40) Patel, R. A.; Webb, M. A. Data-Driven Design of Polymer-Based Biomaterials: High-throughput Simulation, Experimentation, and Machine Learning. *ACS Applied Bio Materials* **2024**, *7*, 510–527, DOI: 10.1021/acsabm.2c00962.
- (41) Gallagher, Q. M.; Webb, M. A. Data efficiency of classification strategies for chemical and materials design. *Digital Discovery* **2025**, *4*, 135–148, DOI: 10.1039/D4DD00298A.
- (42) An, Y.; Webb, M. A.; Jacobs, W. M. Active learning of the thermodynamics-dynamics trade-off in protein condensates. *Science Advances* **2024**, *10*, eadj2448, DOI: 10.1126/sciadv.adj2448.
- (43) Tamasi, M. J.; Patel, R. A.; Borca, C. H.; Kosuri, S.; Mugnier, H.; Upadhyaya, R.; Murthy, N. S.; Webb, M. A.; Gormley, A. J. Machine Learning on a Robotic Platform for the Design of Polymer–Protein Hybrids (Adv. Mater. 30/2022). *Advanced Materials* **2022**, *34*, DOI: 10.1002/adma.202270219.
- (44) Kim, C.; Chandrasekaran, A.; Jha, A.; Ramprasad, R. Active-learning and materials design: the example of high glass transition temperature polymers. *MRS Communications* **2019**, *9*, 860–866, DOI: 10.1557/mrc.2019.78.
- (45) Zhao, W.; Fu, X.; Xu, X.; Zhang, L.; Wang, L.; Lin, J.; Hu, Y.; Gao, L.; Du, L.; Tian, X. Design of multicomponent thermosetting polymers with enhanced tensile properties through active learning. *Composites Science and Technology* **2024**, *256*, 110779, DOI: 10.1016/j.compscitech.2024.110779.

-
- (46) Zhou, H.; Fang, Y.; Gao, H. Using Active Learning for the Computational Design of Polymer Molecular Weight Distributions. *ACS Engineering Au* **2024**, *4*, 231–240, DOI: 10.1021/acsengineeringau.3c00056.
- (47) Chang, W.-C.; Tsai, Z.-Y.; Chen, C.-W.; Yu, C.-H.; Chen, C.-S. Mechanical property prediction of random copolymers using uncertainty-based active learning. *Computational Materials Science* **2025**, *247*, 113489, DOI: 10.1016/j.commatsci.2024.113489.
- (48) Patel, R. A.; Borca, C.; Webb, M. Featurization Strategies for Polymer Sequence or Composition Design by Machine Learning. *Molecular Systems Design & Engineering* **2021**, *7*, 661–676, DOI: 10.1039/D1ME00160D.
- (49) Webb, M. A.; Jackson, N.; Gil, P. S.; Pablo, J. D. d. Targeted sequence design within the coarse-grained polymer genome. *Science Advances* **2020**, *6*, DOI: 10.1126/sciadv.abc6216.
- (50) Khabaz, F.; Khare, R. Effect of chain architecture on the size, shape, and intrinsic viscosity of chains in polymer solutions: a molecular simulation study. *The Journal of Chemical Physics* **2014**, *141*, 214904, DOI: 10.1063/1.4902052.
- (51) Wijesinghe, S.; Perahia, D.; Grest, G. S. Polymer Topology Effects on Dynamics of Comb Polymer Melts. *Macromolecules* **2018**, *51*, 7621–7628, DOI: 10.1021/acs.macromol.8b01449.
- (52) Wever, D. A. Z.; Polgar, L. M.; Stuart, M. C. A.; Picchioni, F.; Broekhuis, A. A. Polymer Molecular Architecture As a Tool for Controlling the Rheological Properties of Aqueous Polyacrylamide Solutions for Enhanced Oil Recovery. *Industrial & Engineering Chemistry Research* **2013**, *52*, 16993–17005, DOI: 10.1021/ie403045y.
- (53) Fan, Z. W.; Jin, X. L.; Chen, Y.; Lu, M.; Wang, Y. R.; Yue, K.; Wen, T.; Tang, L.; Wu, Z. L.; Sun, T. Topology and Dynamic Regulations of Comb-like Polymers as Strong Adhesives. *Macromolecules* **2023**, *56*, 1514–1526, DOI: 10.1021/acs.macromol.2c01851.

-
- (54) Xiong, C.; Xiong, W.; Mu, Y.; Pei, D.; Wan, X. Mussel-inspired polymeric coatings with the antifouling efficacy controlled by topologies. *Journal of Materials Chemistry B* **2022**, *10*, 9295–9304, DOI: 10.1039/D2TB01851A.
- (55) Modica, K. J.; Martin, T. B.; Jayaraman, A. Effect of Polymer Architecture on the Structure and Interactions of Polymer Grafted Particles: Theory and Simulations. *Macromolecules* **2017**, *50*, 4854–4866, DOI: 10.1021/acs.macromol.7b00524.
- (56) Webb, M. A.; Jung, Y.; Pesko, D. M.; Savoie, B. M.; Yamamoto, U.; Coates, G. W.; Balsara, N. P.; Wang, Z.; Miller, T. F. Systematic Computational and Experimental Investigation of Lithium-Ion Transport Mechanisms in Polyester-Based Polymer Electrolytes. *ACS Central Science* **2015**, *1*, 198–205, DOI: 10.1021/acscentsci.5b00195.
- (57) Polymeropoulos, G.; Zapsas, G.; Ntetsikas, K.; Bilalis, P.; Gnanou, Y.; Hadjichristidis, N. 50th Anniversary Perspective: Polymers with Complex Architectures. *Macromolecules* **2017**, *50*, 1253–1290, DOI: 10.1021/acs.macromol.6b02569.
- (58) Aoki, Y.; Wu, S.; Tsurimoto, T.; Hayashi, Y.; Minami, S.; Okubo, T.; Shiratori, K.; Yoshida, R. Multitask Machine Learning to Predict Polymer–Solvent Miscibility Using Flory–Huggins Interaction Parameters. *Macromolecules* **2023**, *56*, 5446–5456, DOI: 10.1021/acs.macromol.2c02600.
- (59) Ethier, J. G.; Audus, D. J.; Ryan, D. C.; Vaia, R. A. Integrating theory with machine learning for predicting polymer solution phase behavior. *Giant* **2023**, *15*, 100171, DOI: 10.1016/j.giant.2023.100171.
- (60) Jiang, S.; Webb, M. A. Physics-Guided Neural Networks for Transferable Property Prediction in Architecturally Diverse Copolymers. *Macromolecules* **2025**, *58*, 4971–4984, DOI: 10.1021/acs.macromol.5c00720.
- (61) Audus, D. J.; McDannald, A.; DeCost, B. Leveraging Theory for En-

-
- hanced Machine Learning. *ACS Macro Letters* **2022**, *11*, 1117–1122, DOI: 10.1021/acsmacrolett.2c00369.
- (62) Bejagam, K. K.; Lalonde, J.; Iverson, C. N.; Marrone, B. L.; Pilania, G. Machine Learning for Melting Temperature Predictions and Design in Polyhydroxyalkanoate-Based Biopolymers. *The Journal of Physical Chemistry B* **2022**, *126*, 934–945, DOI: 10.1021/acs.jpcc.1c08354.
- (63) Chew, A. K.; Sender, M.; Kaplan, Z.; Chandrasekaran, A.; Elk, J. C.; Browning, A. R.; Kwak, H. S.; Halls, M. D.; Afzal, M. A. F. Advancing material property prediction: using physics-informed machine learning models for viscosity. *Journal of Cheminformatics* **2024**, *16*, DOI: 10.1186/s13321-024-00820-5.
- (64) Himanshu; Chakraborty, K.; Patra, T. K. Developing efficient deep learning model for predicting copolymer properties. *Physical Chemistry Chemical Physics* **2023**, *25*, 25166–25176, DOI: 10.1039/D3CP03100D.
- (65) Tran, H. D.; Kim, C.; Chen, L.; Chandrasekaran, A.; Batra, R.; Venkatram, S.; Kamal, D.; Lightstone, J. P.; Gurnani, R.; Shetty, P.; Ramprasad, M.; Laws, J.; Shelton, M.; Ramprasad, R. Machine-learning predictions of polymer properties with Polymer Genome. *Journal of Applied Physics* **2020**, *128*, DOI: 10.1063/5.0023759.
- (66) Mysona, J. A.; Nealey, P. F.; Pablo, J. J. d. Machine Learning Models and Dimensionality Reduction for Prediction of Polymer Properties. *Macromolecules* **2024**, *57*, 1988–1997, DOI: 10.1021/acs.macromol.3c02401.
- (67) Sánchez-Lengeling, B.; Aspuru-Guzik, A. Inverse molecular design using machine learning: Generative models for matter engineering. *Science* **2018**, *361*, 360–365, DOI: 10.1126/science.aat2663.
- (68) Goldberg, D. E. *Genetic Algorithms in Search, Optimization and Machine Learning*, 1st ed.; Addison-Wesley Longman Publishing Co., Inc.: USA, 1989.

-
- (69) Kneiding, H.; Balcells, D. Augmenting genetic algorithms with machine learning for inverse molecular design. *Chemical Science* **2024**, *15*, 15522–15539, DOI: 10.1039/d4sc02934h.
- (70) Carreau, P. J. Rheological Equations from Molecular Network Theories. *Transactions of the Society of Rheology* **1972**, *16*, 99–127, DOI: 10.1122/1.549276.
- (71) Yasuda, K.; Armstrong, R. C.; Cohen, R. E. Shear flow properties of concentrated solutions of linear and star branched polystyrenes. *Rheologica Acta* **1981**, *20*, 163–178, DOI: 10.1007/bf01513059.
- (72) Jouenne, S.; Levache, B. Universal viscosifying behavior of acrylamide-based polymers used in enhanced oil recovery. *Journal of Rheology* **2020**, *64*, 1295–1313, DOI: 10.1122/8.0000063.
- (73) Schroeder, C. M.; Teixeira, R. E.; Shaqfeh, E. S. G.; Chu, S. Dynamics of DNA in the Flow-Gradient Plane of Steady Shear Flow: Observations and Simulations. *Macromolecules* **2005**, *38*, 1967–1978, DOI: 10.1021/ma0480796.
- (74) Karniadakis, G. E.; Kevrekidis, I. G.; Lu, L.; Perdikaris, P.; Wang, S.; Yang, L. Physics-informed machine learning. *Nature Reviews Physics* **2021**, *3*, 422–440, DOI: 10.1038/s42254-021-00314-5.
- (75) Jin, J.; Pak, A. J.; Durumeric, A. E. P.; Loose, T. D.; Voth, G. A. Bottom-up Coarse-Graining: Principles and Perspectives. *Journal of Chemical Theory and Computation* **2022**, *18*, 5759–5791, DOI: 10.1021/acs.jctc.2c00643.
- (76) Dhamankar, S.; Webb, M. A. Chemically specific coarse-graining of polymers: Methods and prospects. *Journal of Polymer Science* **2021**, 1–31, DOI: 10.1002/pol.20210555.
- (77) Noid, W. G. Perspective: Advances, Challenges, and Insight for Predictive Coarse-

Grained Models. *The Journal of Physical Chemistry B* **2023**, *127*, 4174–4207, DOI: 10.1021/acs.jpcc.2c08731.

- (78) Venkatram, S.; Kim, C.; Chandrasekaran, A.; Ramprasad, R. Critical Assessment of the Hildebrand and Hansen Solubility Parameters for Polymers. *Journal of Chemical Information and Modeling* **2019**, *59*, 4188–4194, DOI: 10.1021/acs.jcim.9b00656.
- (79) Mathieu, D. Pencil and Paper Estimation of Hansen Solubility Parameters. *ACS Omega* **2018**, *3*, 17049–17056, DOI: 10.1021/acsomega.8b02601.
- (80) McCleary-Petersen, K. C.; Wiegand, K. R.; Taleff, M. T.; Guironnet, D. Engineering Polymer Architecture Through Reaction Rates. *Macromolecules* **2024**, *58*, 18–31, DOI: 10.1021/acs.macromol.4c01662.
- (81) Lin, T.-S.; Coley, C. W.; Mochigase, H.; Beech, H. K.; Wang, W.; Wang, Z.; Woods, E.; Craig, S. L.; Johnson, J. A.; Kalow, J. A.; Jensen, K. F.; Olsen, B. D. BigSMILES: A Structurally-Based Line Notation for Describing Macromolecules. *ACS Central Science* **2019**, *5*, 1523–1531, DOI: 10.1021/acscentsci.9b00476.
- (82) Gao, H.; Oakley, L. H.; Konstantinov, I. A.; Arturo, S. G.; Broadbelt, L. J. Acceleration of Kinetic Monte Carlo Method for the Simulation of Free Radical Copolymerization through Scaling. *Industrial & Engineering Chemistry Research* **2015**, *54*, 11975–11985, DOI: 10.1021/acs.iecr.5b03198.
- (83) Mavrantzas, V. G. Using Monte Carlo to Simulate Complex Polymer Systems: Recent Progress and Outlook. *Frontiers in Physics* **2021**, *9*, DOI: 10.3389/fphy.2021.661367.
- (84) Agarwal, U.; Doelder, J. d. Integrated Kinetic Monte Carlo–Structure–Rheology Model for Solution Copolymerization of Ethylene and α -Olefins. *Macromolecules* **2019**, *52*, 293–310, DOI: 10.1021/acs.macromol.8b02000.

-
- (85) Stubbs, J.; Carrier, R.; Sundberg, D. C. Monte Carlo Simulation of Emulsion Polymerization Kinetics and the Evolution of Latex Particle Morphology and Polymer Chain Architecture. *Macromolecular Theory and Simulations* **2008**, *17*, 147–162, DOI: 10.1002/mats.200700061.
- (86) Anderson, J. A.; Glaser, J.; Glotzer, S. C. HOOMD-blue: A Python package for high-performance molecular dynamics and hard particle Monte Carlo simulations. *Computational Materials Science* **2020**, *173*, 109363, DOI: 10.1016/j.commatsci.2019.109363.
- (87) Noguchi, H.; Kikuchi, N.; Gompper, G. Particle-based mesoscale hydrodynamic techniques. *EPL (Europhysics Letters)* **2007**, *78*, 10005, DOI: 10.1209/0295-5075/78/10005.
- (88) Howard, M. P.; Panagiotopoulos, A. Z.; Nikoubashman, A. Efficient Mesoscale Hydrodynamics: Multiparticle Collision Dynamics with Massively Parallel GPU Acceleration. *Computer Physics Communications* **2018**, *230*, 10–20, DOI: 10.1016/j.cpc.2018.04.009.
- (89) Müller-Plathe, F. Reversing the perturbation in nonequilibrium molecular dynamics: An easy way to calculate the shear viscosity of fluids. *Physical Review E* **1999**, *59*, 4894–4898, DOI: 10.1103/physreve.59.4894.
- (90) Petersen, M. K.; Lechman, J. B.; Plimpton, S. J.; Grest, G. S.; in 't Veld, P. J.; Schunk, P. R. Mesoscale hydrodynamics via stochastic rotation dynamics: Comparison with Lennard-Jones fluid. *The Journal of Chemical Physics* **2010**, *132*, 174106, DOI: 10.1063/1.3419070.
- (91) Ashbaugh, H. S.; Hatch, H. W. Natively Unfolded Protein Stability as a Coil-to-Globule Transition in Charge/Hydrophathy Space. *Journal of the American Chemical Society* **2008**, *130*, 9536–9542, DOI: 10.1021/ja802124e.pdf.pdf.

-
- (92) Bishop, M.; Kalos, M. H.; Frisch, H. L. Molecular dynamics of polymeric systems. *The Journal of Chemical Physics* **1979**, *70*, 1299–1304, DOI: 10.1063/1.437567.
- (93) Niepert, M.; Ahmed, M.; Kutzkov, K. Learning Convolutional Neural Networks for Graphs. Proceedings of The 33rd International Conference on Machine Learning. New York, New York, USA, 2016; pp 2014–2023.
- (94) Grattarola, D.; Alippi, C. Graph Neural Networks in TensorFlow and Keras with Spektral. *IEEE Computational Intelligence Magazine* **2021**, *16*, 99–106, DOI: 10.1109/mci.2020.3039072.
- (95) Xu, K.; Hu, W.; Leskovec, J.; Jegelka, S. How Powerful are Graph Neural Networks? International Conference on Learning Representations. 2018.
- (96) Abadi, M. et al. TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems. 2015; <https://www.tensorflow.org/>, Software available from tensorflow.org.
- (97) Kingma, D. P.; Ba, J. Adam: A Method for Stochastic Optimization. *arXiv preprint arXiv:1412.6980* **2014**,
- (98) Gonzalez, T. F. Clustering to minimize the maximum intercluster distance. *Theoretical Computer Science* **1985**, *38*, 293–306, DOI: 10.1016/0304-3975(85)90224-5.
- (99) Johnson, M. E.; Moore, L. M.; Ylvisaker, D. Minimax and maximin distance designs. *Journal of Statistical Planning and Inference* **1990**, *26*, 131–148, DOI: 10.1016/0378-3758(90)90122-b.
- (100) Branch, M. A.; Coleman, T. F.; Li, Y. A Subspace, Interior, and Conjugate Gradient Method for Large-Scale Bound-Constrained Minimization Problems. *SIAM Journal on Scientific Computing* **1999**, *21*, 1–23, DOI: 10.1137/s1064827595289108.
- (101) Williams, C. K.; Rasmussen, C. E. *Gaussian processes for machine learning*; MIT press Cambridge, MA, 2006; Vol. 2.

-
- (102) Jones, D. R.; Schonlau, M.; Welch, W. J. Efficient Global Optimization of Expensive Black-Box Functions. *Journal of Global Optimization* **1998**, *13*, 455–492, DOI: 10.1023/a:1008306431147.
- (103) Uhrenholt, A. K.; Jensen, B. S. Efficient Bayesian Optimization for Target Vector Estimation. International Conference on Artificial Intelligence and Statistics (AISTATS). 2019; pp 2661–2670.
- (104) Zhu, C.; Byrd, R. H.; Lu, P.; Nocedal, J. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software (TOMS)* **1997**, *23*, 550–560, DOI: 10.1145/279232.279236.
- (105) Vatin, M.; Kundu, S.; Locatelli, E. Conformation and dynamics of partially active linear polymers. *Soft Matter* **2024**, *20*, 1892–1904, DOI: 10.1039/D3SM01162C.
- (106) Lundberg, S. M.; Lee, S.-I. A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems (NeurIPS). 2017; pp 4765–4774, DOI: 10.5555/3295222.3295230.

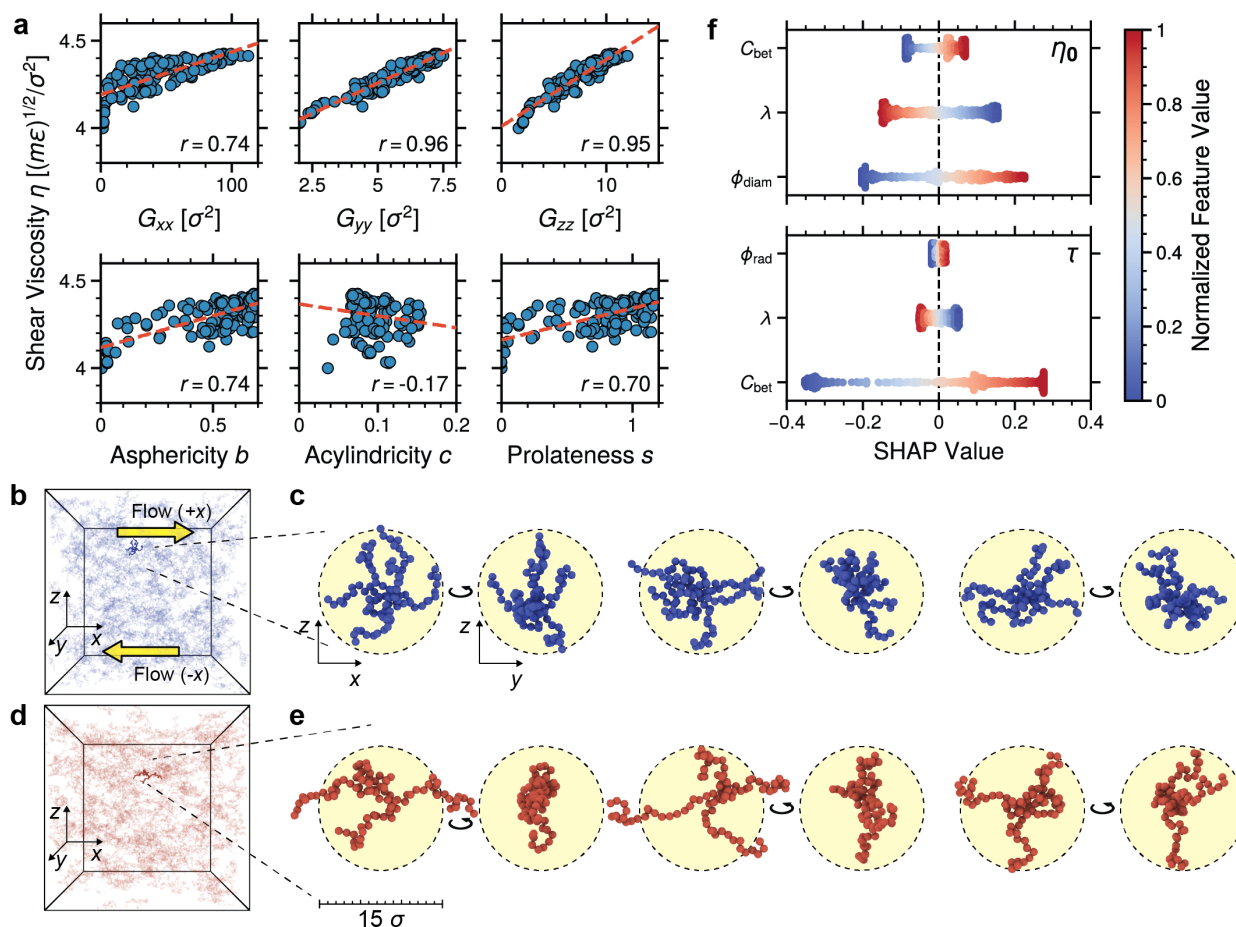


Fig. 6: Topological and conformational features associated with shear viscosity. **a**, Shear viscosity plotted against gyration tensor components (G_{xx} , G_{yy} , G_{zz}) and shape descriptors (asphericity b , acylindricity c , prolateness s), computed under constant shear. Pearson correlation coefficients (r) of the linear fits (red dashed) are shown. **b–e**, Simulation snapshots of individual polymer chains at three time points, spaced by $10^3 \tau$, under steady shear. In each pair, the same polymer conformation is shown from two orthogonal perspectives: the left view lies in the xz plane, parallel to the flow direction, and the right view is obtained by rotation (arrow) into the yz plane, transverse to the flow. The highlighted chain is overlaid with circles of diameter 15σ as a visual reference for conformational spread. Other chains are rendered semi-transparent. Two representative systems are shown: **b–c**, $\lambda = 0.1578$; **d–e**, $\lambda = 0.3383$. Both are simulated under identical shear stress and exhibit similar steady-state shear viscosities, despite differences in solvophobicity and conformation. **f**, SHAP values for Lasso models predicting zero-shear viscosity η_0 and relaxation time τ . Color denotes normalized input feature values. Positive SHAP values indicate a positive contribution to the predicted rheological variable.